

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5
Duration :60 Minutes

Total Marks: 40

Code of Conduct

I **G.Hima Bindu(192511377)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G.Hima Bindu*

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
- Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time

field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Retail Data Optimization Problem

Introduction

A retail company generates data from physical stores, online shopping, inventory systems, mobile applications, and customer interactions. A big data architecture can integrate all these sources to provide real-time inventory tracking, demand forecasting, and personalized recommendations.

Proposed Big Data Architecture

1. Data Sources

- Point-of-sale (POS) systems
- E-commerce website and mobile application
- Inventory and warehouse systems
- Customer browsing and purchase history
- Social media and customer feedback

2. Data Ingestion

- Apache Kafka can collect real-time sales and customer events.
- Apache NiFi can be used to collect and transfer data from different systems.
- Batch data can be loaded using ETL tools.

3. Data Storage

- HDFS or cloud object storage can store large volumes of raw data.
- Data Lake stores structured, semi-structured, and unstructured data.
- Data Warehouse stores processed data for reporting and business analysis.

4. Data Processing

- Apache Spark can process large datasets quickly.
- Spark Streaming can process real-time sales and inventory events.
- Data can be cleaned, transformed, and integrated before analysis.

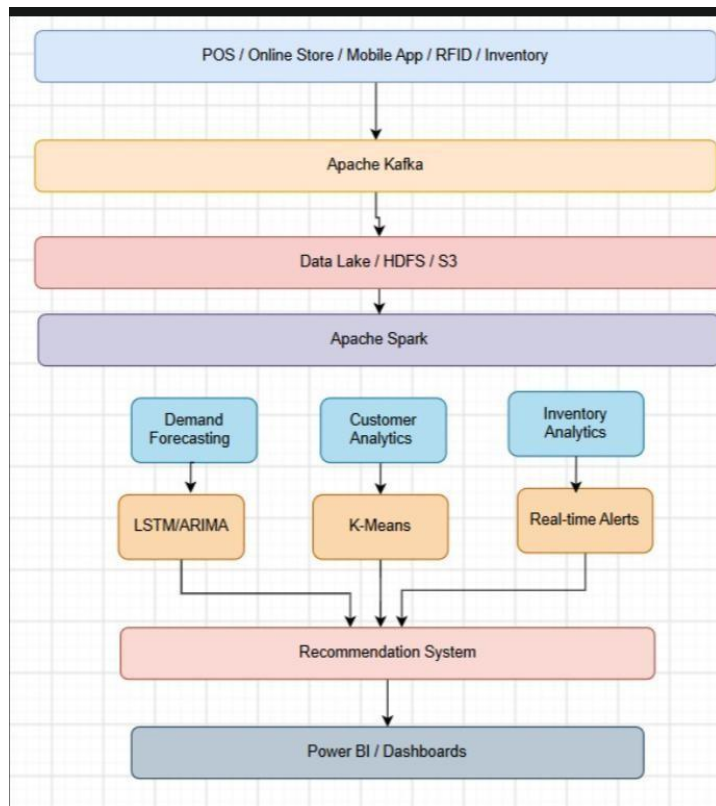
5. Analytics and Machine Learning

- **Demand forecasting:** Time-series models such as ARIMA, Prophet, or LSTM can predict future product demand.
- **Recommendation system:** Collaborative filtering and content-based filtering can recommend products.
- **Customer segmentation:** K-Means clustering can group customers based on purchasing behavior.

6. Real-Time Inventory

When a product is sold, the transaction is sent through Kafka and processed by Spark Streaming. The inventory database is immediately updated. Alerts can be generated when stock reaches a minimum level.

Architecture Flow



This architecture provides real-time visibility of inventory, improves demand forecasting, reduces stock-outs, and enables personalized marketing. Big data technologies also help the retailer make faster and more accurate business decisions.

2.Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Social Media Fraud Detection Problem

Introduction

A social media platform generates billions of posts, comments, likes, shares, and user interactions. A scalable big data system is required to identify fake accounts and misinformation with minimum delay.

Proposed Big Data Solution

1. Data Sources

- Posts and comments
- Likes and shares
- User profiles
- Login and activity information
- Friend/follower relationships
- Reported content

2. Stream Data Ingestion

- Apache Kafka can collect billions of user events continuously.
- Kafka provides high throughput and fault tolerance.

3. Stream Processing

- Apache Flink **or** Spark Streaming processes events in real time.
- It can identify suspicious behavior immediately.

For example, if a new account creates thousands of posts and sends requests to many users within a short period, the system can generate a fraud alert.

4. Machine Learning

Machine learning models can identify patterns that indicate fake accounts or misinformation.

- **Logistic Regression:** Classifies suspicious users or content.
- **Random Forest:** Detects complex fraud patterns.

- **Gradient Boosting:** Provides accurate classification.
- **Neural Networks:** Can analyze complex user and content patterns.
- **Graph algorithms:** Detect suspicious relationships between accounts.

5. Feature Extraction

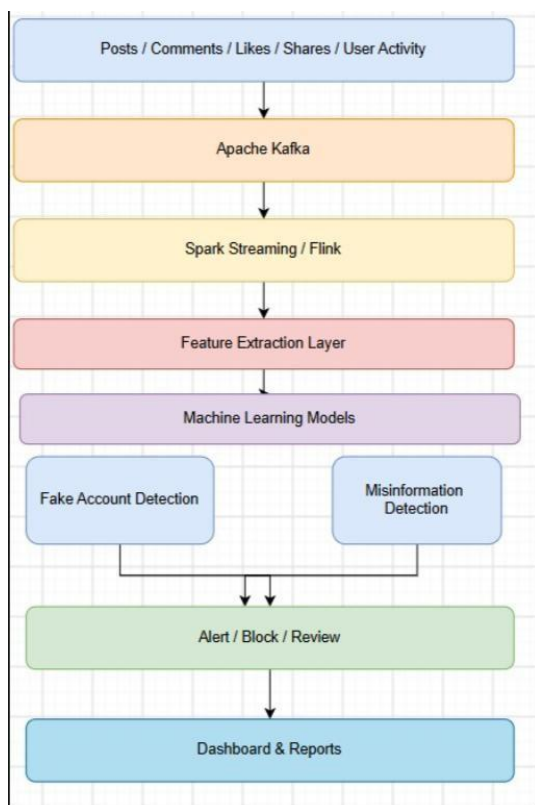
Important features include:

- Account age
- Number of followers and following
- Posting frequency
- IP/device patterns
- Sharing behavior
- Text similarity
- Network relationships

6. Storage

Historical data can be stored in a data lake using HDFS or cloud storage. Databases such as NoSQL systems can store large-scale user activity data.

Architecture Flow



Combining stream processing with machine learning allows the platform to detect suspicious activity quickly. Kafka provides scalable data ingestion, while Flink or Spark provides real-time processing. ML models improve detection accuracy as more historical data becomes available.

3. Government Data Platform Problem A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Government Data Platform Problem

Introduction

A national government platform may contain census, economic, geographical, healthcare, and administrative data. Since multiple departments need to share this information, proper data governance, interoperability, privacy, and security are essential.

Proposed Governance and Data Management Strategy

1. Central Data Lake

A government data lake can store data from different departments in a common platform.

- Census data
- Economic data
- Geospatial data
- Public service data
- Administrative records

2. Data Integration

ETL/ELT tools can extract data from different departments, transform it into standard formats, and load it into the common platform.

3. Interoperability

Common standards should be established for:

- Data formats
- Metadata
- APIs
- Data definitions
- Geographic coordinates
- Identification systems

This ensures that different government departments can understand and exchange data correctly.

4. Data Governance

A data governance framework should define:

- Data ownership
- Data quality standards
- Data access policies
- Data retention rules
- Data classification
- Responsibility of departments

5. Security

Security mechanisms include:

- Role-Based Access Control (RBAC)
- Multi-factor authentication
- Encryption at rest and in transit
- Network security
- Audit logs
- Regular security monitoring

6. Privacy Protection

Sensitive citizen information should be protected using:

- Data masking
- Anonymization
- Pseudonymization
- Access restrictions

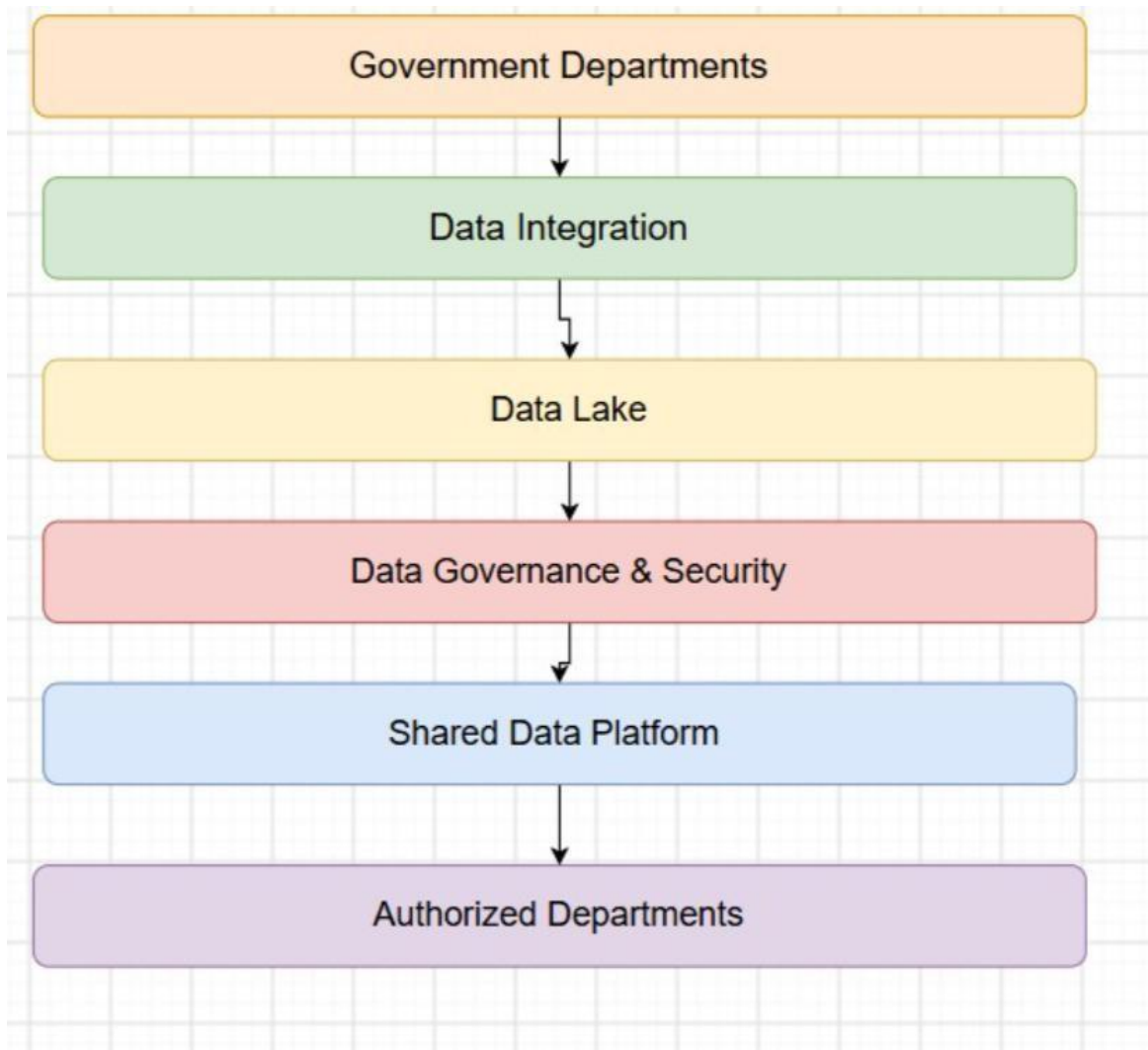
Only authorized users should access sensitive information.

7. Data Quality

Automated validation should identify:

- Duplicate records
- Missing values
- Incorrect formats
- Inconsistent information

Architecture Flow



A centralized and governed data platform improves data sharing between departments while maintaining privacy and security. Standardization and strong governance ensure interoperability, data quality, and responsible use of government data.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Banking Fraud Detection Problem

Introduction

Banks process millions of transactions every minute. Fraud detection must analyze both historical and real-time transaction data. The system should provide low-latency detection while maintaining high accuracy.

Proposed Big Data Pipeline

1. Data Sources

- Credit/debit card transactions
- Online banking transactions
- ATM transactions
- Mobile banking
- Customer account information
- Historical fraud records

2. Real-Time Data Ingestion

Apache Kafka can collect transaction events continuously. It supports high-throughput and fault-tolerant data streaming.

3. Stream Processing

Apache Flink or Spark Streaming can process transactions in real time.

The system can check:

- Transaction amount
- Location
- Transaction frequency
- Device information
- Customer spending pattern
- Previous fraudulent behavior

4. Feature Engineering

Features can include:

- Average transaction amount
- Number of transactions per hour
- Distance between transactions
- New device usage
- Unusual transaction location
- Frequency of failed transactions

5. Fraud Detection Algorithms

- **Logistic Regression:** Basic fraud classification.
- **Random Forest:** Handles multiple transaction features.
- **XGBoost:** High-performance classification.
- **Isolation Forest:** Detects unusual transactions.
- **Neural Networks:** Detect complex fraud patterns.

A combination of rule-based detection and machine learning can improve accuracy.

6. Historical Data

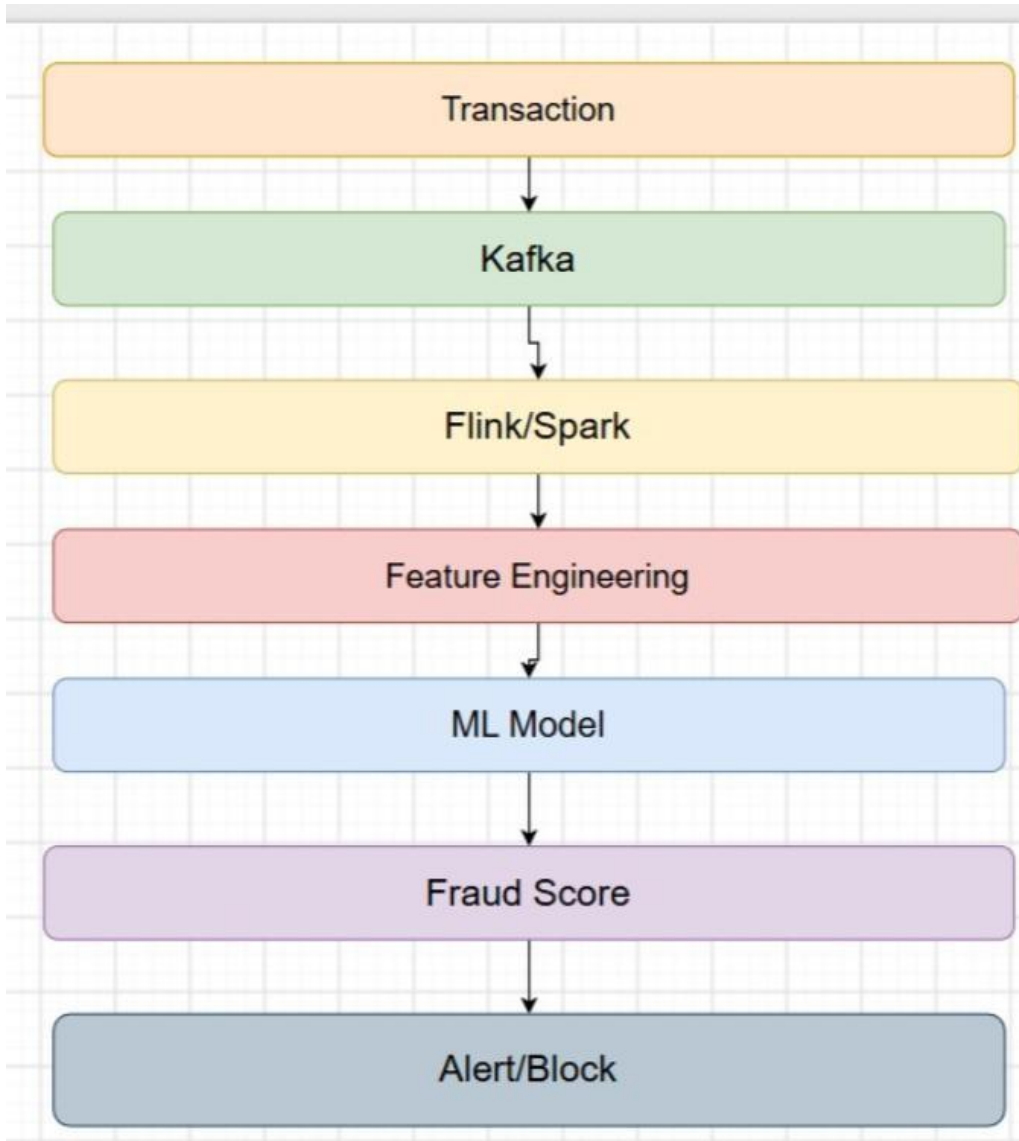
Historical transactions and fraud records can be stored in a data lake or data warehouse. These records are used to train and update ML models.

7. Real-Time Alert

If a transaction receives a high fraud score, the system can:

- Block the transaction
- Ask for additional authentication
- Send an alert to the customer
- Notify the fraud investigation team

Architecture Flow



A streaming big data architecture provides fast fraud detection with low latency. Kafka handles continuous transaction streams, while Flink or Spark performs real-time processing. Machine learning improves the detection of complex and previously unknown fraud patterns.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Agriculture Smart Farming Problem

Introduction

Precision agriculture uses sensors, drones, satellites, weather stations, and farm equipment to collect large amounts of data. Big data analytics can convert this information into useful insights for improving crop yield and reducing resource consumption.

Proposed Big Data Solution

1. Data Sources

- Soil moisture sensors
- Temperature sensors
- Humidity sensors
- Weather stations
- Drones
- Satellite images
- Crop monitoring systems
- Farm equipment

2. Data Collection

IoT sensors continuously collect field information. MQTT or Apache Kafka can be used to transfer sensor data to the processing system.

3. Data Storage

A data lake can store:

- Sensor data
- Drone images
- Satellite images
- Weather information
- Historical crop data

HDFS or cloud object storage can be used for large-scale storage.

4. Data Processing

Apache Spark can process large amounts of sensor and image data. Streaming tools can process real-time sensor information.

5. Analytics

Different analytics techniques can support farming decisions:

- **Predictive analytics:** Predict crop yield and disease risk.
- **Weather analysis:** Predict irrigation and harvesting conditions.
- **Machine learning:** Identify crop diseases and estimate yield.
- **Image analysis:** Analyze drone images to identify unhealthy crops.
- **Clustering:** Divide fields into zones based on soil and crop conditions.

6. Precision Irrigation

If sensors detect low soil moisture, the system can automatically recommend or activate irrigation for that specific area instead of watering the entire field.

7. Decision Support

A dashboard can display:

- Soil moisture
- Weather conditions
- Crop health
- Irrigation requirements
- Disease alerts
- Expected crop yield

Architecture Flow

Sensors/Drones/Satellites → IoT Gateway → Kafka/MQTT → Data Lake → Spark Processing → ML Analytics → Farmer Dashboard/Automated Actions

Big data enables farmers to make data-driven decisions. Real-time sensor analysis helps optimize water, fertilizer, and pesticide usage, while machine learning and image analytics can improve crop health and yield. Thus, precision agriculture increases productivity while reducing resource wastage.